

An Infinite-Dimensional Banach Algebra With A Nonzero Hyper-Commutator

Automatic math-discovery smoke test

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Source Problem

The source paper is Maysam Maysami Sadr, *A Remark on Contractible Banach Algebras of Operators*, arXiv:2211.03493 [1]. The paper defines a hyper-commutator for a Banach algebra A to be an element $M \in A \widehat{\otimes}_\gamma A$ such that

$$aM = Ma \quad (a \in A),$$

where A acts on $A \widehat{\otimes}_\gamma A$ by multiplication on the left and right tensor factors. Immediately after the definition, the paper asks:

Does there exist an infinite-dimensional Banach algebra with a nonzero hyper-commutator?

that M is a hyper-commutator for A if

$$aM = Ma \quad (a \in A).$$

By definition, diagonals of contractible Banach algebras are hyper-commutator elements. Following the discussion of §1, the next question is very natural.

QUESTION 3.6. *Does there exist an infinite-dimensional Banach algebra with a nonzero hyper-commutator?*

The next theorem which is the main result of this note, establishes a null-property of hyper-commutators.

Result Type

This packet gives a **candidate full affirmative answer** to the question as stated. In fact the example can be chosen unital.

Proof Intuition

The fastest way to make the two tensor-module actions agree is to put a large square-zero ideal I inside a unital algebra. If $I^2 = 0$, then elements of I cannot see the I -part of an algebra element when multiplied on either side; only the scalar part of the unitization remains. Therefore every elementary tensor $u \otimes v$ with $u, v \in I$ is scaled by the same scalar on the left and on the right. Choosing u and v nonzero gives a nonzero hyper-commutator.

The Example

Lemma 1. *Let E be a Banach space. On the Banach space*

$$A = \mathbb{C} \oplus_1 E, \quad \|(\lambda, x)\| = |\lambda| + \|x\|,$$

define

$$(\lambda, x)(\mu, y) = (\lambda\mu, \lambda y + \mu x).$$

Then A is a unital Banach algebra with identity $(1, 0)$. The closed ideal $I = \{0\} \oplus E$ satisfies $I^2 = 0$.

Proof. This is the usual unitization of the Banach algebra E equipped with the zero product. Associativity follows directly from associativity of scalar multiplication. The element $(1, 0)$ is a two-sided identity. Finally,

$$\begin{aligned} \|(\lambda, x)(\mu, y)\| &= |\lambda\mu| + \|\lambda y + \mu x\| \\ &\leq |\lambda||\mu| + |\lambda| \|y\| + |\mu| \|x\| \\ &\leq (|\lambda| + \|x\|)(|\mu| + \|y\|), \end{aligned}$$

so multiplication is jointly continuous and submultiplicative for the ℓ^1 -sum norm. If $x, y \in E$, then

$$(0, x)(0, y) = (0, 0),$$

so $I^2 = 0$. □

Theorem 1. *There exists an infinite-dimensional unital Banach algebra with a nonzero hyper-commutator.*

Proof. Let E be any infinite-dimensional Banach space, for instance $E = \ell^1$, and form the Banach algebra $A = \mathbb{C} \oplus_1 E$ from Lemma 1. Choose nonzero vectors $u, v \in E$, and set

$$p = (0, u), \quad q = (0, v), \quad M = p \otimes q \in A \widehat{\otimes}_\gamma A.$$

The projective tensor norm is a cross norm, so

$$\|p \otimes q\|_\gamma = \|p\| \|q\| > 0.$$

Hence $M \neq 0$ in $A \widehat{\otimes}_\gamma A$.

Let $a = (\lambda, x) \in A$. Since $I^2 = 0$,

$$ap = (\lambda, x)(0, u) = (0, \lambda u) = \lambda p, \quad qa = (0, v)(\lambda, x) = (0, \lambda v) = \lambda q.$$

Using the standard outer bimodule actions on the projective tensor product,

$$aM = (ap) \otimes q = \lambda(p \otimes q) = p \otimes (\lambda q) = p \otimes (qa) = Ma.$$

Thus M is a nonzero hyper-commutator for the infinite-dimensional Banach algebra A . □

Corollary 1. *The question quoted above from arXiv:2211.03493 has an affirmative answer.*

Scope And Limitations

The construction is intentionally elementary and answers the literal question as written. It does not address stronger variants in which one also requires the algebra to be semisimple, to be represented faithfully as an operator algebra containing finite-rank operators, to have nonzero image under a specific canonical operator-tensor map, or to be contractible.

Verification Notes

The only points needing verification are:

1. $A = \mathbb{C} \oplus_1 E$ is a Banach algebra. This is checked in Lemma 1.
2. $p \otimes q \neq 0$ in the completed projective tensor product. This follows from the cross-norm identity for elementary tensors, or equivalently by applying bounded functionals detecting p and q .
3. The hyper-commutator identity uses only $I^2 = 0$, so it holds for every $a \in A$.

The accompanying script `code/check_square_zero_hypercommutator.py` checks the displayed left/right action identities in a finite-dimensional square-zero model.

Novelty And Literature Check

Before promotion, the run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2211.03493, the source title, *hyper-commutator*, *nonzero hyper-commutator*, and related phrases. No duplicate packet, prior attempt, or proof-gap entry answering this question was found.

A bounded web search on June 29, 2026 for exact phrases including *nonzero hyper-commutator*, *hyper-commutator Banach algebra*, and the full question found the source arXiv page but no separate later answer. The local parsed-source corpus likewise found the hyper-commutator question in the source paper but no later explicit resolution.

References

- [1] Maysam Maysami Sadr, *A Remark on Contractible Banach Algebras of Operators*, arXiv:2211.03493, 2022.